

Optimization 2

ISE 5406

Lecture 11: Steepest Descent Method and Line Search

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Last Time (Lecture 10): Optimality Conditions and Line Search Foundations

- First-order necessary conditions: $\nabla f(\mathbf{x}^*) = \mathbf{0}$
- Second-order conditions (necessary and sufficient)
- Line search: motivation and 1D methods

Today (Lecture 11): Steepest Descent Method, Line Search, and Convergence

- Steepest descent for multivariate problems
- Inexact line search: Armijo and Wolfe conditions
- Convergence analysis: L-smoothness and strong convexity
- Condition numbers and iteration complexity

Next Topics

- [Lecture 12](#): Newton's method and quasi-Newton methods
- [Lecture 13](#): Conjugate gradient methods
- [Lecture 14](#): BFGS and other advanced first-order methods

Big Picture: Today we master gradient descent through the lens of descent directions and line search, then develop convergence theory that explains both **why** gradient descent works and **how fast** it converges.

Outline

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

Descent Directions

Definition (Descent Direction)

A direction $\mathbf{d} \in \mathbb{R}^n$ is a **descent direction** at $\mathbf{x}$ if $\nabla f(\mathbf{x})^T \mathbf{d} < 0$.

Intuition: Moving from $\mathbf{x}$ in direction $\mathbf{d}$ (with small step) decreases the function.

Descent Direction Theorem

Let f be continuously differentiable at $\mathbf{x}$ with $\nabla f(\mathbf{x}) \neq \mathbf{0}$. Then $\mathbf{d}$ is a descent direction iff:

$$\nabla f(\mathbf{x})^T \mathbf{d} < 0.$$

Examples:

- $\mathbf{d} = -\nabla f(\mathbf{x})$ always satisfies $\nabla f(\mathbf{x})^T \mathbf{d} = -\|\nabla f(\mathbf{x})\|^2 < 0$
(Steepest descent)
- $\mathbf{d} = -P\nabla f(\mathbf{x})$ where $P \succ 0$ satisfies
 $\nabla f(\mathbf{x})^T \mathbf{d} = -\nabla f(\mathbf{x})^T P \nabla f(\mathbf{x}) < 0$ (Preconditioned)

Steepest Descent Direction

Lemma: Steepest Descent Direction

For any $\mathbf{x}$ where $\nabla f(\mathbf{x}) \neq 0$, the solution to

$$\min_{\|\mathbf{d}\|=1} \nabla f(\mathbf{x})^T \mathbf{d}$$

is

$$\mathbf{d}^* = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}.$$

The minimum value is $-\|\nabla f(\mathbf{x})\|$.

Proof: By Cauchy-Schwarz: $|\nabla f(\mathbf{x})^T \mathbf{d}| \leq \|\nabla f(\mathbf{x})\| \|\mathbf{d}\|$. With $\|\mathbf{d}\| = 1$, equality holds when $\mathbf{d}$ is proportional to $-\nabla f(\mathbf{x})$.

Interpretation: The negative gradient is the direction that decreases f fastest (in terms of directional derivative).

In Practice

We use $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$ directly (not normalized). This scales the step size automatically via line search.

Steepest Descent Algorithm

Algorithm: Steepest Descent with Line Search

Initialization: Choose starting point $\mathbf{x}_1 \in \mathbb{R}^n$, tolerance $\epsilon > 0$, set

$k = 1$.

1. **Compute gradient:** $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$
2. **Termination test:** If $\|\nabla f(\mathbf{x}_k)\| < \epsilon$, **stop**.
3. **Line search:** Compute $\lambda_k > 0$ such that $f(\mathbf{x}_k + \lambda_k \mathbf{d}_k)$ is reduced (e.g., via backtracking)
4. **Update:** $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$
5. **Iterate:** $k \leftarrow k + 1$, return to step 1

Key features:

- **Guaranteed descent:** $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$ (if line search is done correctly)
- **No Hessian needed:** Only gradients required
- **Potential zig-zag:** On quadratics, can exhibit slow zigzag convergence

Steepest Descent on Quadratics

Consider a **convex quadratic**:

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x} + d$$

where $Q \succ 0$. The gradient is $\nabla f(\mathbf{x}) = Q\mathbf{x} + \mathbf{c}$.

For steepest descent, $\mathbf{d}_k = -\nabla f(\mathbf{x}_k) = -Q\mathbf{x}_k - \mathbf{c}$.

Exact line search (optimal λ_k): The 1D minimization gives:

$$\lambda_k^* = \frac{\|\mathbf{d}_k\|^2}{\mathbf{d}_k^T Q \mathbf{d}_k} = \frac{\|\nabla f(\mathbf{x}_k)\|^2}{\nabla f(\mathbf{x}_k)^T Q \nabla f(\mathbf{x}_k)}.$$

Key property: Consecutive gradients are orthogonal under exact line search:

$$\nabla f(\mathbf{x}_{k+1})^T \nabla f(\mathbf{x}_k) = 0.$$

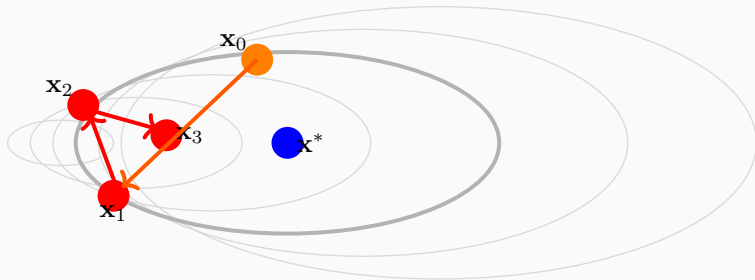
Convergence: For quadratics with eigenvalues

$0 < \lambda_{\min}(Q) \leq \dots \leq \lambda_{\max}(Q)$, define condition number:

$$\kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

The convergence factor is $\rho = 1 - \frac{1}{\kappa(Q)}$, so $\|\mathbf{x}_k - \mathbf{x}^*\| \leq \rho^k \|\mathbf{x}_0 - \mathbf{x}^*\|$.

Zigzag Behavior on Quadratics



Observation: Steepest descent on elongated elliptical level sets (high condition number) exhibits a **zigzag pattern**.

Why? Consecutive gradient directions are orthogonal, so iterates bounce back and forth across the minor axis instead of moving smoothly toward the optimum.

Numerical Example: Steepest Descent

Example

Consider $f(x_1, x_2) = x_1^2 + 4x_2^2$ with starting point $\mathbf{x}_0 = (4, 2)^T$.

1. Compute the first iterate using exact line search.
2. Show that consecutive gradients are orthogonal.

Solution:

Step 1: Compute gradient at $\mathbf{x}_0$: $\nabla f(\mathbf{x}_0) = (2x_1, 8x_2)|_{(4,2)} = (8, 16)^T$.

Step 2: Descent direction: $\mathbf{d}_0 = -(8, 16)^T = (-8, -16)^T$. We have $\|\mathbf{d}_0\|^2 = 64 + 256 = 320$.

Step 3: Quadratic form: For $Q = \text{diag}(2, 8)$,

$$\mathbf{d}_0^T Q \mathbf{d}_0 = 64 \cdot 2 + 256 \cdot 8 = 128 + 2048 = 2176.$$

Step 4: Exact step size:

$$\lambda_0 = \frac{320}{2176} = \frac{5}{34}.$$

Step 5: Next iterate:

$$\mathbf{x}_1 = (4, 2) + \frac{5}{34}(-8, -16) = (4 - \frac{40}{34}, 2 - \frac{80}{34}) = (\frac{76}{17}, \frac{-12}{17}).$$

Step 6: Check orthogonality: $\nabla f(\mathbf{x}_1) = (2 \cdot \frac{76}{17}, 8 \cdot \frac{-12}{17}) = (\frac{152}{17}, \frac{-96}{17})$.

$$\nabla f(\mathbf{x}_1)^T \nabla f(\mathbf{x}_0) = \frac{152}{17} \cdot 8 + \frac{-96}{17} \cdot 16 = \frac{1216 - 1536}{17} = \frac{-320}{17} \neq 0.$$

Note: Orthogonality holds in the *Q-weighted sense*, not the standard dot product.

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

Motivation: Inexact Line Search

Exact line search: Minimize $\theta(\lambda) = f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ over $\lambda > 0$.

Problems:

- Expensive: requires many function evaluations per iteration
- Often unnecessary: we just need **sufficient progress**, not the exact minimum
- Overkill: in later iterations, approximate minimizers are fine

Solution: Accept any λ_k that satisfies the **Wolfe conditions**:

- **Sufficient decrease (Armijo)** condition: guarantee progress
- **Curvature condition**: avoid tiny steps

Benefit

Inexact line search reduces overhead while maintaining global convergence guarantees!

Wolfe Conditions

Let $c_1, c_2 \in (0, 1)$ with $c_1 < c_2$. A step size $\lambda > 0$ satisfies the **Wolfe conditions** if:

Armijo (Sufficient Decrease):

$$f(\mathbf{x}_k + \lambda \mathbf{d}_k) \leq f(\mathbf{x}_k) + c_1 \lambda \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$$

Curvature Condition:

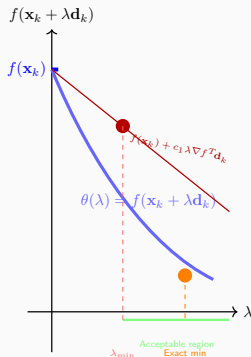
$$\nabla f(\mathbf{x}_k + \lambda \mathbf{d}_k)^T \mathbf{d}_k \geq c_2 \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$$

Interpretation:

- **Armijo**: Actual function value must drop by at least a fraction c_1 of the predicted decrease
- **Curvature**: Slope at new point is less steep (in magnitude) than at starting point, preventing tiny steps

Practical values: $c_1 = 10^{-4}$ (very small), $c_2 = 0.9$ (curvature check is loose).

Armijo Condition: Geometric Picture



Key observations:

- **Blue curve:** Actual function $\theta(\lambda) = f(\mathbf{x}_k + \lambda \mathbf{d}_k)$
- **Red line:** Armijo upper bound (much less steep since $c_1 \ll 1$)
- **Acceptable:** Any λ where $\theta(\lambda)$ lies **below** the red line
- **Efficient:** Accept first λ that satisfies Armijo (e.g., backtracking)

Backtracking Line Search Algorithm

Algorithm: Backtracking Line Search

Input: $\mathbf{x}_k$, descent direction $\mathbf{d}_k$, parameters $c_1 \in (0, 0.5)$, $\rho \in (0, 1)$.

1. Start with $\lambda = 1$
2. **While** $f(\mathbf{x}_k + \lambda \mathbf{d}_k) > f(\mathbf{x}_k) + c_1 \lambda \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$ **do**:
 - Set $\lambda \leftarrow \rho \lambda$
3. **Return** λ

Properties:

- **Simple:** Only requires gradient at $\mathbf{x}_k$ and function values at trial points
- **Guaranteed termination:** For descent direction $\mathbf{d}_k$, Armijo will be satisfied for sufficiently small λ
- **Practical:** Usually converges in 3–5 iterations

Note: This implements **Armijo only**. Strong Wolfe (both conditions) requires more sophisticated algorithms like the **zoom procedure**.

Backtracking Example

Exercise: Backtracking Line Search

Let $f(x_1, x_2) = e^{x_1} + e^{x_2} + x_1^2 + x_2^2$. At $\mathbf{x}_k = (0, 0)^T$, use steepest descent direction $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$.

With $c_1 = 10^{-4}$ and $\rho = 0.5$, find the step size λ by backtracking.

Solution

Step 1: Gradient: $\nabla f(\mathbf{x}_k) = (e^0 + 2 \cdot 0, e^0 + 2 \cdot 0) = (1, 1)^T$ at $(0, 0)$.

Step 2: Descent direction: $\mathbf{d}_k = -(1, 1)^T$. Verify:

$$\nabla f^T \mathbf{d}_k = (1, 1)^T \cdot (-1, -1)^T = -2 < 0 \quad \checkmark$$

Step 3: Check $\lambda = 1$:

- $f(\mathbf{x}_k + 1 \cdot \mathbf{d}_k) = f(-1, -1) = e^{-1} + e^{-1} + 1 + 1 = 2e^{-1} + 2 \approx 2.736$
- $f(\mathbf{x}_k) + c_1 \cdot 1 \cdot (-2) = 2 + 10^{-4}(-2) = 1.9998$
- Since $2.736 > 1.9998$, Armijo **fails**.

Step 4: Backtrack: $\lambda = 0.5$. Check:

- $f(-0.5, -0.5) = e^{-0.5} + e^{-0.5} + 0.25 + 0.25 \approx 1.213 + 0.5 = 1.713$
- Armijo RHS: $2 + 10^{-4}(0.5)(-2) = 1.9999$
- Since $1.713 < 1.9999$, Armijo **holds** $\checkmark$. Return $\lambda = 0.5$.

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

L -Smooth Functions: Definition

Definition (L -Smoothness)

A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth if ∇f is L -Lipschitz:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

Equivalent characterizations:

- **Hessian:** If f is twice differentiable, then f is L -smooth iff $\nabla^2 f(\mathbf{x}) \preceq LI_n$ for all $\mathbf{x}$.
- **Function value:** f is L -smooth iff for all $\mathbf{x}, \mathbf{y}$,

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

Intuition: The function is “sandwiched” between its tangent line and a quadratic. Smoothness rules out functions with kinks (like $|x|$).

Example: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x}$ with $Q \preceq LI$ is L -smooth.

The Descent Lemma

Theorem (Descent Lemma)

For an L -smooth function f , if gradient descent uses step size

$\lambda \leq 1/L$:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{\lambda}{2} \|\nabla f(\mathbf{x}_k)\|^2.$$

Proof sketch: By L -smoothness at $\mathbf{y} = \mathbf{x}_k - \lambda \nabla f(\mathbf{x}_k)$:

$$\begin{aligned} f(\mathbf{x}_{k+1}) &\leq f(\mathbf{x}_k) - \lambda \|\nabla f(\mathbf{x}_k)\|^2 + \frac{L\lambda^2}{2} \|\nabla f(\mathbf{x}_k)\|^2 \\ &= f(\mathbf{x}_k) - \lambda \left(1 - \frac{L\lambda}{2}\right) \|\nabla f(\mathbf{x}_k)\|^2 \\ &\leq f(\mathbf{x}_k) - \frac{\lambda}{2} \|\nabla f(\mathbf{x}_k)\|^2 \end{aligned}$$

where the last step uses $\lambda \leq 1/L$, so $1 - L\lambda/2 \geq 1/2$.

Each step makes guaranteed progress proportional to $\|\nabla f(\mathbf{x}_k)\|^2$.

Convergence of GD on Convex, L -Smooth Functions

Theorem ($O(1/T)$ Convergence Rate)

If f is convex and L -smooth, and we use $\lambda = 1/L$:

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{L\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2T}.$$

Interpretation:

- To achieve $f(\mathbf{x}_T) - f^* \leq \varepsilon$, we need $T = O(1/\varepsilon)$ iterations
- **Sublinear rate:** Progress slows down over time
- **Disappointing:** To double accuracy, must double iterations

Proof sketch:

1. By convexity: $f(\mathbf{x}^*) \geq f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T(\mathbf{x}^* - \mathbf{x}_k)$
2. Rearrange: $\nabla f(\mathbf{x}_k)^T(\mathbf{x}_k - \mathbf{x}^*) \geq f(\mathbf{x}_k) - f(\mathbf{x}^*)$
3. Combine with descent lemma
4. Telescope over T steps

Key takeaway: Convexity + smoothness alone is **not enough** for fast convergence. We need strong convexity!

μ -Strongly Convex Functions

Definition (Strong Convexity)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex if for all $\mathbf{x}, \mathbf{y}$:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

Equivalent forms:

- **Hessian:** $\nabla^2 f(\mathbf{x}) \succeq \mu I_n$ for all $\mathbf{x}$
- **Quadratic lower bound:** $f(\mathbf{x}) - \frac{\mu}{2} \|\mathbf{x}\|^2$ is convex
- **Monotone gradients:** $(\nabla f(\mathbf{x}) - \nabla f(\mathbf{y}))^T (\mathbf{x} - \mathbf{y}) \geq \mu \|\mathbf{x} - \mathbf{y}\|^2$

Examples:

- **(yes)** $f(\mathbf{x}) = a\|\mathbf{x}\|^2$, $a > 0$ is $2a$ -strongly convex
- **(yes)** $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$, $Q \succ 0$ is $2\lambda_{\min}(Q)$ -strongly convex
- **(no)** $f(x) = -\log x$ is NOT strongly convex

Intuition: Function is bounded **below** by a quadratic “bowl”. This gives fast convergence!

The Polyak–Łojasiewicz Inequality

Theorem (Polyak–Łojasiewicz (PL) Inequality)

If f is μ -strongly convex:

$$\|\nabla f(\mathbf{x})\|^2 \geq 2\mu(f(\mathbf{x}) - f(\mathbf{x}^*)) \quad \forall \mathbf{x}.$$

Proof: By strong convexity at $\mathbf{y} = \mathbf{x}^*$ with $\nabla f(\mathbf{x}^*) = 0$:

$$f(\mathbf{x}) - f(\mathbf{x}^*) \geq \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|^2.$$

Also by convexity:

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \nabla f(\mathbf{x})^T (\mathbf{x} - \mathbf{x}^*) \leq \|\nabla f(\mathbf{x})\| \cdot \|\mathbf{x} - \mathbf{x}^*\|.$$

Combining and rearranging:

$$\|\nabla f(\mathbf{x})\| \geq \sqrt{2\mu(f(\mathbf{x}) - f(\mathbf{x}^*))}$$

which gives the result.

Large gradient $\Rightarrow$ **large suboptimality gap**. This is the key to exponential convergence!

Linear Convergence: Main Theorem

Theorem (Linear Convergence of GD)

Let f be μ -strongly convex and L -smooth. With $\lambda = 1/L$:

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{\mu}{L}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)).$$

Define condition number $\kappa = L/\mu$. Then:

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{1}{\kappa}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)).$$

This is exponential (linear) convergence!

- Each iteration reduces gap by factor $1 - 1/\kappa$
- For ε -accuracy: $T = O(\kappa \log(1/\varepsilon))$ iterations
- **Compare:** Convex only needed $O(1/\varepsilon)$ iterations!

Proof:

1. **Descent Lemma:** $f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L} \|\nabla f(\mathbf{x}_k)\|^2$
2. **PL Inequality:** $\|\nabla f(\mathbf{x}_k)\|^2 \geq 2\mu(f(\mathbf{x}_k) - f(\mathbf{x}^*))$
3. **Combine:** $f(\mathbf{x}_{k+1}) - f^* \leq (1 - \mu/L)(f(\mathbf{x}_k) - f^*)$
4. **Recurse:** Telescoping gives the result

Iterate Convergence (Corollary)

Corollary (Iterate Convergence)

Under the same assumptions:

$$\|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq \left(1 - \frac{1}{\kappa}\right)^{k-1} \|\mathbf{x}_1 - \mathbf{x}^*\|^2.$$

Key steps:

- Strong convexity gives: $f(\mathbf{x}_k) - f^* \geq \frac{\mu}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2$
- Smoothness gives: $f(\mathbf{x}_1) - f^* \leq \frac{L}{2} \|\mathbf{x}_1 - \mathbf{x}^*\|^2$
- Combine with functional convergence bound

Remark: The leading factor κ can be absorbed; the main convergence behavior is the geometric rate $(1 - 1/\kappa)^k$.

The Condition Number

Definition (Condition Number)

For an L -smooth, μ -strongly convex function:

$$\kappa = \frac{L}{\mu} \geq 1.$$

For a quadratic $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x}$ with $Q \succ 0$:

$$\kappa = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

Convergence in terms of κ :

$$\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq \left(1 - \frac{1}{\kappa}\right)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Iteration complexity: For $\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq \varepsilon \|\mathbf{x}_0 - \mathbf{x}^*\|^2$:

$$T \geq \kappa \ln(1/\varepsilon).$$

A large condition number κ means slow convergence! This explains the zigzag behavior observed earlier.

Geometric Picture

- Well-conditioned (κ small): nearly circular level sets $\Rightarrow$ fast convergence
- Ill-conditioned (κ large): elongated elliptical level sets $\Rightarrow$ zigzagging

Condition Number Examples on Quadratics

For a quadratic with exact line search, the convergence factor is:

$$\rho = \frac{\kappa - 1}{\kappa + 1}, \quad \text{giving} \quad \|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq \rho^{2k} \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Examples

- $Q = \text{diag}(0.5, 1)$: $\kappa = 2$, convergence factor $\rho = 1/3$. **Fast!**
- $Q = \text{diag}(0.01, 1)$: $\kappa = 100$, convergence factor $\rho = 99/101 \approx 0.98$.
Slow!
- $Q = \text{diag}(10^{-6}, 1)$: $\kappa = 10^6$, convergence factor $\rho \approx 1 - \varepsilon$. **Very slow!**

Summary table:

Assumption	Rate	Iterations for ε -accuracy
Convex + L -smooth	$O(1/T)$	$O(1/\varepsilon)$
+ μ -strongly convex	$O((1 - \mu/L)^T)$	$O(\kappa \log(1/\varepsilon))$

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

Zoutendijk's Theorem

Theorem (Zoutendijk (Global Convergence))

Suppose f is continuously differentiable, $\{x_k\}$ remains in a compact set, and each step satisfies the Wolfe conditions with $c_1 \in (0, 1/2)$. Then:

$$\sum_{k=1}^{\infty} \cos^2(\theta_k) \|\nabla f(\mathbf{x}_k)\|^2 < \infty,$$

where θ_k is the angle between $\mathbf{d}_k$ and $-\nabla f(\mathbf{x}_k)$.

Consequence for steepest descent ($\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$, so $\theta_k = 0$, $\cos(\theta_k) = 1$):

$$\sum_{k=1}^{\infty} \|\nabla f(\mathbf{x}_k)\|^2 < \infty.$$

This implies: Either

- (i) The algorithm terminates at a stationary point, or
- (ii) $\|\nabla f(\mathbf{x}_k)\| \rightarrow 0$.

Practical Implication

Steepest descent with Wolfe line search is **globally convergent** on any continuously differentiable function with bounded level sets!

Steepest Descent Method

Line Search Methods

Convergence Theory: Smoothness and Strong Convexity

$\kappa = L/\mu$ governs speed

Ill-conditioned problems require **advanced methods** (Newton, CG, BFGS)

Looking Ahead: Next Lecture

Lecture 12: Newton's Method

- **Newton's method:** Use Hessian information for faster convergence
- **Quadratic convergence:** Exponentially fast near the optimum
- **Trade-off:** More expensive per iteration, but fewer iterations
- **Modifications:** Damping strategies and line search for global convergence
- **Motivation:** Why we need quasi-Newton methods (BFGS, L-BFGS)

Why Newton's Method?

- Steepest descent: $O(\kappa \log(1/\varepsilon))$ iterations for ε -accuracy
- Newton: $O(\log \log(1/\varepsilon))$ iterations (superfast near solution!)
- But Newton requires solving linear systems: $\nabla^2 f(\mathbf{x}_k) \mathbf{d}_k = -\nabla f(\mathbf{x}_k)$
- Quasi-Newton approximates Hessian to reduce cost per iteration

Bridge the gap between simple (gradient descent) and powerful (Newton) methods!